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B. E. (Fourth Semester) Examination, Nov.-Dec. 2016

(New Scheme)

(Mech. Branch)

APPLIED THERMODYNAMICS

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

***Note** : All questions are compulsory. Part (a) of each question is compulsory. Attempt any two parts from (b), (c) and (d). Use steam table is permitted.*

Unit - I

1. (a) Arrange in decending order, the efficiencies of Auto, Diesel and Dual cycle for a constant maximum pressure and temperature? 2

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(b) Derive the equation for the thermal efficiency and mean effective pressure for an otto cycle. 7

(c) A diesel engine has a compression ratio of 14 and cut-off takes place at 6% of the stroke. Find the air standard efficiency. 7

(d) Prove that, for minimum temperature T_1 and maximum temperature T_3, the efficiency of otto cycle is given by

$$\eta_{otto} = 1 - \left(\frac{T_{min}}{T_{max}}\right)^{1/2}$$ 7

Unit - II

2. (a) What are the different types of compressor? 2

(b) Define and derive the volumetric efficiency of a compressor. 7

(c) A single-acting two stage air compressor deals with 4 m^3/min of air at 1.013 bar and 15°C with a speed of 250 rpm. The delivery pressure is 80 bar. Assuming complete intercooling, find the minimum power required by the compressor and the bore and

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stroke of the compressor. Assume a piston speed of 3 m/sec mechanical efficiency of 75% and volumetric efficiency of 80% per stage. Assume the polytropic index of compression in both the stage to be n = 1·25 and neglect clearance. 7

(d) A three-stage single acting reciprocating air compressor has a low-pressure (L.P.) cylinder of 450 mm bore and 300 mm stroke. The clearance volume of the L.P. cylinder is 5% of the swept volume. Intake pressure and temperature are 1 bar and 18°C respectively. While the delivery pressure is 15 bar. Intermediate pressure are ideal and intercooling is perfect. The compression and expansion indexd can be taken as 1·3.

Estimate :

(i) The intermediate pressure

(ii) The effective swept volume

(iii) Temperature and volume of air delivered per stroke at 15 bar

(iv) The work done per kg of air.

Take R = 0·29 kJ/kg K 7

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Unit - III

3. (a) Define Rankine cycle. 2

(b) Explain Binary vapour cycle. 7

(c) A steam power station uses the following cycle :

Steam at boiler outlet - 150 bar, 550°C

Reheat at 40 bar to 550°C

Condensor at 0.1 bar

Using Mollier chart and assuming ideal processes, find the :

(i) Quality of steam at turbine exhaust

(ii) Cycle efficiency

(iii) Steam rate 7

(d) A steam turbine working on Rankine cycle is supplied with dry saturated steam at 25 bar and the exhaust takes place at 0.2 bar. For the steam flow rate at 10 Kg/sec.

Determine :

(i) Quality of steam at the end of expansion

(ii) Turbine shaft work

(iii) Power required to drive the pump

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(iv) Work ratio

(v) Rankine efficiency

(vi) Heat flow in the condenser. 7

Unit - IV

4. (a) Define Condensers. 2

(b) Explain the classification and working principle of cooling tower. 7

(c) The following observation were made on a steam condensing plant :

Barometer reading = 760 mm of Hg.

Recorded vacuum = 700 mm of Hg.

Mean temperature of condensation = 34°C

Hotwell temperature = 27°C

Mass of condensate = 2120 kg/hr.

Mass of cooling water = 66000 kg/hr.

Rise in temperature of cooling water = 16°C

Find :

(i) The state of steam entering the condenser.

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(ii) Mass of air present in kg/m^3 of condenser.

(iii) Vacuum efficiency. 7

(d) Steam enters a jet condenser having an absolute pessure of 0.16 bar with a dryness fraction of 0·88 and is condensed by water at 12°C. Determine the quality of water required per kg of steam and the temperature of condensate and cooling water having the condenser. Assume that the air present is 1/3 by mass that of steam. Gas equation for dry air and for the steam are $pV = 0{\cdot}03$ T and $pV = 0{\cdot}048$ T respectively, where p is in p_a and V is in m^3/kg. 7

Unit - V

5. (a) Define Mach Number. 2

(b) Derive the equation

$$\frac{dA}{A} = \left(m^2 - 1\right)\frac{dV}{V}$$ 7

(c) A steam of air flows in a duct of 100 mm diameter at a rate of 1 kg/sec. The stagnation temperature is 37°C. At one section of the duct the static pressure is 40 kPa. Calculate Mach number, velocity and stegnation pressure at this section. 7

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(d) Air at $p_0 = 1{\cdot}4$ bar, $T_0 = 280$ K expands isentropically through a converging nozzle and discharge to the atmosphere at 1 bar. The exit plane area is $0{\cdot}0013$ m^3. Determine the mass flow rate. Is it possible to increase the mass flow rate by increasing the supply pressure? 7

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